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Studierichting: Informatica
Oefeningenreeks 4A nr. 36.14

$$\int \frac{4dx}{2x^2+7}$$
$$= \int \frac{4dx}{7\left(\frac{2}{7}x^2+1\right)}$$
$$u = \sqrt{\frac{2}{7}}x$$
$$du = \sqrt{\frac{2}{7}}dx$$
$$\frac{du}{\sqrt{\frac{2}{7}}} = dx$$
$$\Rightarrow \frac{4}{7} \cdot \frac{1}{\sqrt{\frac{2}{7}}} \int \frac{1}{u^2+1}du$$
$$= \frac{4}{7\sqrt{\frac{2}{7}}} \operatorname{Bgtan}\left(\sqrt{\frac{2}{7}}x\right) + C$$
$$= \frac{4}{7\sqrt{\frac{2}{7}}} \cdot \frac{\sqrt{\frac{2}{7}}}{\sqrt{\frac{2}{7}}} \operatorname{Bgtan}\left(\sqrt{\frac{2}{7}}x\right) + C$$
$$= 2\sqrt{\frac{2}{7}} \operatorname{Bgtan}\left(\sqrt{\frac{2}{7}}x\right) + C$$

# References